Cyclic cores and high-type two-group polyhedra

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Abstract

We recover the all-parameter existence theorem and the complete eight-family high-type classification of Hou, Feng and Leemans. Existence follows by attaching two dihedral quotients to a common affine square group. For classification, a cyclic-core bound reduces every candidate to one of three bounded square quotients. Explicit extension models give all orders and markings; converse arguments and sharp presentation bounds prove completeness. A single exhaustive order-256 kernel computation supplies the remaining finite seed, without a group catalogue.

1 The source results and common inputs

We recover Theorems 1.2 and 1.3 of Hou, Feng and Leemans [2], in the author edition `arXiv:1802.08947v2` dated 22 January 2019. The first theorem is an existence statement for every admissible rank-three type. The second classifies three high-type families, with two, two and four marked groups respectively. A proof of existence alone does not establish this classification.

We use the marking conventions of the companion common paper [1]. In particular, a string C-group is generated by nonidentity involutions, distant generators commute, and subgroups on two subsets of the generators intersect in the subgroup on their intersection. For rank three we write the marking as (a, b, c), with $ac = ca$. Conjugation is on the right: $x^g = g^{-1}xg$, and $[x, g] = x^{-1}g^{-1}xg$. The subscript of D_{2m} denotes its order. All isomorphisms used to classify the models preserve the ordered marking. The standard correspondence with regular abstract polytopes is retained from [4, Section 2E].

The reusable results used below are UA.Phi, UA.Square, UA.SquareRecognition, UA.Edge and UA.Search, the last including the complete UA.Nilpotent and UA.Ladder proofs. They are fully stated and proved in [1]. The application-specific work consists of the cyclic-core reduction, eight explicit extension models, their converse classification, and recognition of the bare source presentations. The only numerical input is a complete order-256 marked-kernel computation, whose generation is justified by UA.Search. Its full program is included in the shared computational appendix; no abstract-group catalogue or historical classification count is an input.

2 Existence for every stated type

Theorem 1 (Hou–Feng–Leemans, Theorem 1.2)**.** *For integers $n \geq 10$, $s, t \geq 2$ and $s + t \leq n - 1$, there is a rank-three string C-group of order 2^n and type $\{2^s, 2^t\}$.*

Proof. Put $\delta = n - 1 - s - t \geq 0$. The square group $H_{\delta+5}$ of UA.Square has order $2^{\delta+5}$, pair orders $4, 4, 2$, and all Boolean deletion maps. Attach marked dihedral quotients $D_{2^{s+1}} \to D_8$ and $D_{2^{t+1}} \to D_8$ on its two adjacent pairs. Each quotient is injective on its proper marked subgroups, and the supports are the two pairs. UA.Edge gives a string C-group with adjacent orders $2^s, 2^t$ and order

$$2^{\delta+5}\, 2^{s-2}\, 2^{t-2} = 2^n.$$

Distant commutation holds in every coordinate. The retained group–polytope correspondence supplies the regular polyhedron and its full automorphism group. This actually works for $n \geq 5$ whenever the same parameter inequalities hold; the displayed theorem keeps the source's stated range. □

The remaining sections prove the complete classification in Theorem 1.3. For each $n \geq 10$, the types $\{4, 2^{n-3}\}$, $\{4, 2^{n-4}\}$ and $\{4, 2^{n-5}\}$ have exactly two, two and four ordered marked groups. Their complete presentations and all parameter values are given below.

3 Cyclic cores and the eight high-type families

The supported constructions solve existence. Completeness at the high dihedral boundary needs a different local invariant, the cyclic core. Its bounded quotient is nevertheless recognized by the same square model, and its only finite seed is generated by the same UA.Nilpotent/UA.Ladder search as order 1024.

The retained **cyclic core theorem of Lucchini** says: if a cyclic proper subgroup M of a finite group H has core K, then $|M : K| < |H : M|$. Here K is the largest normal subgroup of H contained in M. We use the form recorded in Isaacs, *Finite Group Theory*, Theorem 2.20, also stated as L035 Proposition 2.9.

Theorem H.Core (bounded cyclic quotient). Let $(G; a, b, c)$ be a rank-three string C-group of order 2^{t+s}, type $\{4, 2^t\}$, with $s \geq 3$ and $t \geq s - 2$. Put

$$r = bc, \quad m = 2^t, \quad d_0 = (ab)^2, \quad A = \langle r^{2^{s-2}} \rangle.$$

Then $A \lhd G$, and G/A is a string C-group of order 2^{2s-2} and type $\{4, 2^{s-2}\}$.

Proof. The even-word subgroup $H = \langle ab, bc \rangle$ has index at most two. IP and UA.Phi give $G/G' = C_2^3$, where its image has dimension two, so the index is exactly two. For $M = \langle r \rangle$, the cyclic core theorem gives $|M : \mathrm{Core}_H M| < 2^{s-1}$, hence at most 2^{s-2}. Thus A is the subgroup of the cyclic core of its prescribed order, so is characteristic there and normal in H. The involution b inverts r; consequently $A \lhd G$. It meets $\langle a, b \rangle$ trivially by IP and lies in $\langle b, c \rangle$. Their quotient parabolics still intersect in $\langle bA \rangle$: equality of cosets gives membership in the original right parabolic, after multiplication by an element of A. The left parabolic embeds and the right has its stated dihedral order. This proves IP, independence, type and order in the quotient. ■

Since $\mathrm{Aut}(A)$ is abelian, every commutator centralizes A. In particular $[d_0, r^{2^{s-2}}] = 1$. If $A \neq 1$, its unique involution $z = r^{m/2}$ is central in G. The universal identities

$$r^a = d_0 r, \quad d_0^a = d_0, \quad r^c = r^{-1}, \quad d_0^c = d_0^r \tag{A-C1}$$

follow from the involution, string and order-four relations.

Hereafter $s \in \{3, 4, 5\}$, $n = t + s \geq 10$, so m is divisible by 32. The following eight models will realize every class. Let z be a distinguished central involution of the finite group D:

Branch	Group D	Right-conjugation automorphism T, fixing z
$s = 3$	$\langle d_0, z \rangle = C_2^2$	$T(d_0) = d_0 z^\epsilon$
$s = 4$	$\langle d_0, d_1, z \rangle = C_2^3$	$T(d_0) = d_1,\ T(d_1) = d_0 z^\epsilon$
$s = 5$	Four involutions d_0, d_1, d_2, z, z central, $[d_0, d_2] = 1$, $[d_0, d_1] = [d_1, d_2] = z^\beta$	$T(d_0) = d_1,\ T(d_1) = d_2,\ T(d_2) = d_0 d_1 d_2 z^\nu$

Here $\epsilon \in \{0, 1\}$, or $(\beta, \nu) \in \{0, 1\}^2$. For the last row an explicit group law on four binary coordinates is

$$(u, v, w, e)(u', v', w', e') = (u + u', v + v', w + w', e + e' + \beta(vu' + wv')). \tag{A-C2}$$

Its bilinear final term proves associativity and gives all sixteen normal forms $d_0^u d_1^v d_2^w z^e$. Thus the three D-orders are four, eight and sixteen. The displayed maps preserve their involution and commutator relations and have fourth power one. For example in the last row the two nontrivial commutator

contributions cancel in $(d_0d_1d_2)^2$, and applying T four times to d_0 returns d_0; its orbit generates D together with z.

Adjoin a cyclic r of order m, with $x^r = T(x)$, and take the central quotient

$$N = (D \rtimes \langle r \rangle)/\langle zr^{-m/2} \rangle. \tag{A-C3}$$

The identified subgroup has order two and meets both D and $\langle r \rangle$ trivially, since $T^{m/2} = 1$. Hence D embeds, r retains order m, and $|N| = |D|m/2$.

Define automorphisms $\mathcal{A}, \mathcal{C}$ by

$$\mathcal{A}(r) = d_0 r, \quad \mathcal{A}(d_0) = d_0, \qquad \mathcal{C}(r) = r^{-1}, \quad \mathcal{C}(d_i) = T^{1-i}(d_0), \quad \mathcal{C}(z) = z.$$

In the first two rows $\mathcal{A}$ fixes D; in the third it fixes d_0, d_1, z and sends d_2 to $d_2 z^\beta$. For an explicit verification, substituting the listed generator images into the defining involutions and commutators of D preserves each relation. On its generators the compatibility equalities are

$$\mathcal{A}(Tx) = T(\mathcal{A}(x)^{d_0}), \qquad \mathcal{C}(Tx) = T^{-1}\mathcal{C}(x).$$

They express precisely preservation of $x^r = T(x)$. Collection gives $(d_0 r)^4 = r^4$ in the first two rows and $r^4 z^\nu$ in the last. Since eight divides $m/2$, this preserves $r^m = 1$ and $r^{m/2} = z$. Finally $\mathcal{A}^2 = \mathcal{C}^2 = 1$ and $\mathcal{A}\mathcal{C} = \mathcal{C}\mathcal{A}$ on r, d_0, which generate N. For instance both composites send r to $r^{-1}d_0$. These checks prove the automorphism assertions for every admissible m, not only for a computational sample.

Set $G = N \rtimes \langle a, c \rangle$, where a, c are commuting involutions inducing these automorphisms, and put $b = rc$. Then a, b, c are involutions, $[a, c] = 1$, $bc = r$, and $(ab)^2 = d_0$. They generate G, of order $2^s m$, with pair orders $4, m, 2$. In G/N, the images of a, b are independent whereas the right parabolic has image $\langle cN \rangle$. Hence the only candidates for their intersection are $1, b, d_0, d_0 b$. The normal form (A-C3) gives $d_0 \notin \langle r \rangle$, leaving exactly $\langle b \rangle$. The three generators are independent (already $a \notin \langle b, c \rangle$, and the pair orders exclude the others), so this proves rank-three IP. Direct substitution yields

$$\begin{array}{lll} s = 3: & [d_0, c] = z^\epsilon, & \\ s = 4: & [d_0, r^2] = z^\epsilon, & \\ s = 5: & [a, r^2]^2 = z^\beta, & [a, r^4] = z^\nu. \end{array} \tag{A-C4}$$

They are invariant words of the ordered marking and distinguish the two, two and four models within their branches.

3.1 Completeness and bare presentation recognition

The eight source presentations have common relations

$$a^2 = b^2 = c^2 = (ac)^2 = (ab)^4 = r^m = 1, \qquad r = bc,$$

and the following additional relations, with $z = r^{m/2}$:

Source name	m	Extra relations
G_1, G_2	2^{n-3}	$[d_0, c] = z^\epsilon$, $\epsilon = 0, 1$ respectively
G_3, G_4	2^{n-4}	$[d_0, r^2] = z^\epsilon$, $\epsilon = 0, 1$ respectively
G_5, G_6, G_7, G_8	2^{n-5}	$[a, r^2]^2 = z^\beta$, $[a, r^4] = z^\nu$, with $(\beta, \nu) =$ $(0,0), (1,0), (0,1), (1,1)$

Theorem H.Classification (high-type completeness). Every group in the three classes of L035 Theorem 1.3 is exactly one of the marked models constructed above; each displayed source presentation defines that model.

Proof, first branch. For $s = 3$, Theorem H.Core gives a quotient of type $\{4, 2\}$ and order sixteen, necessarily $D_8 \times C_2$. Thus $h = [d_0, r] \in A = \langle r^2 \rangle$. Since it commutes with r, and d_0 centralizes A, $h^2 = [d_0, r^2] = 1$. Hence $h = 1$ or z, and (A-C1) makes this exactly $[d_0, c] = z^\epsilon$. The conjugates of d_0 obey $d_1 = d_0 z^\epsilon$; (A-C1) forces all the first model's actions and normal forms. The resulting surjection from that model onto the actual group is between groups of the same order, hence an isomorphism.

Second branch. For $s = 4$, the quotient of order 64 and type $\{4, 4\}$ is H_6, by UA.Square Recognition. In that affine quotient $[d_0, r^2] = 1$, so upstairs $h = [d_0, r^2] \in \langle r^4 \rangle$. Again $h^2 = [d_0, r^4] = 1$, giving $d_2 = d_0 z^\epsilon$, where $d_i = d_0^{r^i}$. Conjugation by a fixes d_0, d_1. Applying it to the equation for d_2, with z central, gives $d_2^a = (d_1^{d_0})^r = d_2$, so $[d_0, d_1] = 1$. All second-model relations follow, and the same order comparison proves uniqueness.

The bounded seed for the third branch. The same universal-kernel theorem used for the order-1024 classification gives the following finite fact: every order-256 marked string C-group of type $\{4, 8\}$, with $r = bc, d_0 = (ab)^2$, satisfies exactly one row.

Row	Relation pattern	Ordered marked classes
I	$[a, r^2]^2 = 1,\ [a, r^4] = 1$	1
II	$[a, r^2]^2 = r^4,\ [a, r^4] = 1$	1
III	$[d_0, r^2] = r^4$	1
IV	$[(r^2)^a, r] = r^4$	1

Apply UA.Nilpotent and UA.Ladder, as specialized in UA.Search with d=3 and k=3. The universal class-five group has order 2^{18} and third lower-central subgroup of order 2^{13}. Its invariant index-two ladder has 7, 23 and 61 kernels. Testing all 64 subset pairs for each final candidate gives 37 ordered IP markings, four with the indicated orientation of type {4,8}. The census program in the shared computational appendix evaluates every displayed word on all four and verifies that each belongs to exactly one row, one marking per row. The complete program and every output check are printed in the shared computational appendix; UA.Search supplies the finite-search proof. Thus this bounded input is now conditional on exact GAP/NQ algorithms, with no abstract-group catalogue. No SmallGroups identifier or catalogue multiplicity enters these relation implications.

Third-branch lifts. For $s = 5$, Theorem H.Core gives $A = \langle r^8 \rangle$ and the table applies to G/A. In row III we would have $[d_0, r^2] = r^e$, $e \equiv 4 \pmod 8$. It commutes with r, so $[d_0, r^8] = r^{4e} = 1$, impossible for $m \geq 32$, since $v_2(4e) = 4$. In row IV put $x = (r^2)^a$. The relation would give $[x, r] = r^e$, $e \equiv 4 \pmod 8$, hence $r^x = r^{1-e}$. The unit $1 - e \equiv 5 \pmod 8$ has order $m/4$ modulo m: successive squaring gives $v_2((1-e)^{2^j} - 1) = j + 2$. But $x^{m/8} = (r^{m/4})^a \in A$, by normality, so it centralizes r, contradicting that order. Thus III and IV cannot lift.

In I and II, $[a, r^4] \in A$, so a maps r^4 to an odd power of itself and $M = \langle r^4 \rangle \lhd G$. Quotienting gives a marked $\{4, 4\}$ group of order 128, hence $H_7 = W/4\mathbb{Z}^2$. Let $N = \langle d_0, r \rangle$. Equations (A-C1) make it normal, with index four: modulo N, a, c commute, and their independence follows in the rank-three Frattini quotient, where N has the one-dimensional image spanned by r. The cyclic normal subgroup M is central in N, because d_0 is a commutator and r centralizes its own powers.

In H_7, the conjugates d_i, $0 \leq i \leq 3$, have linear part $-I$ and translation parts $(0, 0), (0, 2), (2, 2)$, $(2, 0)$, up to the choice of coordinate directions. They generate C_2^3, with $d_3 = d_0 d_1 d_2$. Upstairs, every $[d_i, d_j]$ is in central M and has order at most two. Put $[d_0, d_1] = z^\beta$; conjugating by r gives $[d_1, d_2] = z^\beta$. Write $d_3 = d_0 d_1 d_2 u$ with $u \in M$. Conjugation invariance gives

$$[d_1, d_2] = [d_2, d_3] = [d_2, d_0][d_2, d_1],$$

so $[d_0, d_2] = 1$. Consequently $(d_0 d_1 d_2)^2 = 1$; because d_3 is an involution, $u^2 = 1$, giving $u = z^\nu$. One further conjugation gives $d_4 = d_0$. These are precisely the final row of the table of the three model branches, and (A-C1) forces the actions of the commuting complement $\langle a, c\rangle$. The model maps onto G and has its order $32m$, so is isomorphic. This proves completeness of the eight models.

It remains to recognize the **bare presentations**, where the target order cannot yet be assumed. Let P be one of them and $z = r^{m/2}$, initially of order at most two. For G_1, G_2, if $\epsilon = 1$, a centralizes $z = [d_0, c]$ since it centralizes d_0, c; b, c centralize it since they invert r. If $\epsilon = 0$, d_0 is central; (A-C1) and the evenness of $m/2$ then make z central too. Collection gives at most $8m$ normal forms, attained by the first model.

For G_3, G_4, $d_2 = d_0 z^\epsilon$. If $\epsilon = 0$, applying a as in the second-branch argument gives $[d_0, d_1] = 1$, after which the model relations and centrality of z follow. If $\epsilon = 1$, $[d_0, r^2] = z$ implies that conjugation by d_0 inverts z; as $z^2 = 1$, it centralizes z. Thus z commutes with r and all d_i. Write $j = [d_0, d_1]$. Applying a to the equation for d_2, and using (A-C1), gives $z^a = jz$. Since a fixes d_0, d_1, it fixes j, and $a^2 = 1$ gives $j^2 = 1$. Hence d_0, d_1 centralize j. Also $j^r = [d_1, d_2] = [d_1, d_0 z] = j^{-1} = j$, and c exchanges d_0, d_1, so j is central. Collection yields $(d_0 r)^4 = j r^4$. Since $m/8$ is even, $z^a = (d_0 r)^{m/2} = z$, so $j = 1$. The second model relations now give at most $16m$ forms, all attained by that model.

For G_5–G_8, put $w = [a, r^2]$. If $\beta = 1$, conjugation by a inverts w, and so fixes $z = w^2$. If $\nu = 1$, it inverts $[a, r^4] = z$, again fixing z. If both are zero, a centralizes r^4 and therefore z. Thus z is central in every case, and $[a, r^4] = z^\nu$ makes $M = \langle r^4\rangle$ normal of order at most $m/4$. Modulo M, the presentation is W with the extra relation $[a, r^2]^2 = 1$. In the affine model $[a, r^2]$ is a doubled unit translation, so its square and conjugates kill precisely $4\mathbb{Z}^2$. Thus the quotient has order 128, and $|P| \le 128(m/4) = 32m$. The constructed model attains this bound and satisfies every relator, so the order comparison principle identifies P with it. This completes presentation recognition and the theorem. ■

The preceding theorem proves the full marked interpretation of L035 Theorem 1.3; the listed word invariants distinguish all the declared families. It does not classify arbitrary two-groups of order 2^n, other types, or the source's unnumbered exceptional $n < 10$ discussion.

4 Computation, attribution and formal scope

The four shared files `phase78_n009_integration_census.g`, `phase78_n009_integration_finite.g`, `phase78_n009_integration_data.g` and `phase78_n009_integration_run.py` are printed in full in the shared appendix and distributed with this application. The first file generates the order-256 seed independently of the four expected word patterns. It checks every candidate and every subset intersection before testing those words. Its exact output is 37 ordered markings, four of type $\{4, 8\}$, with one in each displayed row. GAP and NQ's exact finite-group algorithms are retained computational machinery. The complete files remain chargeable, including the portions reused by the order-1024 application; no hypothetical smaller checker is counted.

The existence and classification results are due to Hou, Feng and Leemans; the cyclic-core theorem is due to Lucchini. The common supported construction and exhaustive kernel parameter space, together with the explicit extension models and complete presentation bounds, are the present derivation of those results. The calculations have been reused from the reviewed proof bank with all hypotheses retained; no historical priority for the derivation is asserted. The $n < 10$ survey, other types and a classification of arbitrary two-groups are outside this paper's stated contract.

Existing Lean files establish specified kernel and intersection interfaces in the common theory. They do not formalize the cyclic-core instantiation, the eight models, their automorphism calculations, the finite search, or the complete classification theorem. The written proofs and exact computational obligation remain explicit; no new source credit, question answer or length factor is claimed here.

References

[1] Human Compression Project, *Supported kernels, translation lattices and finite marked quotients*, companion common paper, this volume.

[2] D.-D. Hou, Y.-Q. Feng and D. Leemans, *Existence of regular 3-polytopes of order* 2^n, arXiv:1802.08947v2, 22 January 2019; published as *2-groups behaving as automorphism groups of regular 3-polytopes*, Journal of Group Theory 22 (2019), 579–616.

[3] A. Lucchini, *On the order of transitive permutation groups with cyclic point-stabilizer*, Rendiconti Lincei: Matematica e Applicazioni, series 9, 9(4) (1998), 241–243. The cyclic-core form is also recorded in I. M. Isaacs, *Finite Group Theory*, Theorem 2.20, and in [2, Proposition 2.9].

[4] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002.